Fiscal Year:	FY 2022	**Task Last Updated:**	FY 11/01/2021
PI Name:	Tandon, Ritesh Ph.D.		
Project Title:	Effect of Space Radiation on Cytomegalovirus Reactivation and Lytic Replication		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	(1) Immunology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
PI Email:	rtandon@umc.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	601-984-1705
Organization Name:	University of Mississippi Medical Center		
PI Address 1:	Department of Microbiology and Immunology		
PI Address 2:	2500 N State St		
PI Web Page:			
City:	Jackson	**State:**	MS
Zip Code:	39216-4500	**Congressional District:**	3
Comments:			
Project Type:	Ground,New Investigation	**Solicitation / Funding Source:**	2018 Space Biology (ROSBio) NNH18ZTT001N-FG. App B: Flight and Ground Space Biology Research
Start Date:	01/01/2020	**End Date:**	12/31/2022
No. of Post Docs:		**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:	Griko, Yuri	**Contact Phone:**	650-604-0519
Contact Email:	Yuri.V.Griko@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	N/A		
COI Name (Institution):	Crucian, Brian Ph.D. (NASA Johnson Space Center) Mehta, Satish Ph.D. (NASA Johnson Space Center) Simpson, Richard Ph.D. (University of Arizona)		
Grant/Contract No.:	80NSSC19K1603		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Opportunistic pathogens such as latent viruses pose a unique hazard to crew members because of the ineffectiveness of preflight quarantine periods and decreased immune function. Latent herpesviruses reactivate and shed in higher quantities during and after long and short spaceflights. Human cytomegalovirus (CMV) is an important herpesvirus from a public health viewpoint that causes severe systemic infections upon primary infection or on reactivation from latency. The few antiviral drugs available to treat CMV infection suffer from safety, efficacy, and resistance issues, and no vaccine against CMV is currently available. As NASA and collaborating space agencies plan for future exploration class missions (i.e., to Mars or an asteroid), eliminating the risk of an adverse health event due to alterations in virus latency, reactivation, and lytic replication is paramount. Evidence of impaired control of latent viral infections has been documented during short (i.e., 14-day Space Shuttle missions) as well as long (i.e., 6-month International Space Station missions) duration spaceflight missions, with the magnitude of this response being positively associated with time in flight. Shedding of herpesviruses including Epstein–Barr virus (EBV), varicella-zoster virus (VZV), and CMV was detected in >30% of astronauts in a recent study (Mehta et al., 2017) looking at long-duration spaceflight (60–180 days). These viruses reactivated independently of each other and reactivation frequency increased in duration, and amplitude (viral copy numbers) when compared to short duration (10 to 16 days) space shuttle missions. Thus, it is easy to speculate that the substantially longer durations being proposed for exploration class missions (i.e., up to 3 years) will further increase the risk of an adverse health event due to increased virus reactivation and/or replication. Add increased radiation exposure (both duration and magnitude), limited clinical care, and the impossibility of rapidly returning a casualty back to Earth to the list of potential hazards and it is clear to see why obtaining a better understanding of virus replication under space-like conditions is critical to successfully and safely send humans to, and return them from, these remote extraterrestrial destinations. In this proposal, we aim to determine the impact of space radiation on CMV latency, reactivation, and lytic replication. While most studies on this topic in the past have focused on changes in the human immune system that could induce virus reactivation, we will approach this topic from both virus and host standpoints. We will study differences in CMV latency and reactivation before and after radiation exposure and determine if these changes can be correlated with specific changes in virus genome and/or transcriptome. Also, because virus control, dissemination, and resolution are ultimately governed by the competency, or lack thereof, of the host immune system, we will determine the impact of pre-exposing NK-cells and T-cells to space radiation on their ability to control active CMV infections. In this project, we will focus on a single variable: space radiation, rather than looking at multiple variables such as microgravity, physical and psychological stresses of space travel, diet, medications, or other factors associated with spaceflight. The overarching hypothesis is that modeled space radiation will induce specific changes in CMV genome/transcriptome or immune control of infection that will induce CMV reactivation and/or increase in virus replication and pathogenicity. The specific aims to test this hypothesis are: Specific Aim 1: Determine the impact of space radiation on CMV genome and transcriptome. Specific Aim 2: Correlate radiation induced changes in CMV genome and/or transcriptome with CMV latency, reactivation, and lytic replication. Specific Aim 3: Determine the impact of space radiation on immune cell control in a mouse model of CMV infection. Reference: Mehta SK, Laudenslager ML, Stowe RP, Crucian BE, Feiveson AH, Sams CF, Pierson DL. (2017) Latent virus reactivation in astronauts on the international space station, NPJ Microgravity 3, 11.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Upon completion of this project, we will gain a better understanding of how radiation impacts CMV latency, reactivation, and lytic replication. By measuring gene expression on infected cells exposed to radiation, using both traditional and cutting-edge techniques, we will be able to identify CMV genomic and transcriptomic changes that affect virus behavior. We will also determine if space radiation can impair NK-cell and T-cell function, allowing for CMV to replicate and disseminate in the host. This will allow us to determine the risk of CMV infection during spaceflight missions and gain a better understanding of the potential underpinning mechanisms. Findings from this study could lead to the development of appropriate countermeasures to ameliorate potential clinical risks associated with CMV reactivation in flight. Moreover, the methods incorporated here can be adapted to examine the effects of radiation on other relevant viral pathogens known to cause significant disease in immunocompromised people that may also pose a risk to crewmembers during spaceflight missions. Discovering the mechanisms of virus reactivation will help with the development of treatment modalities for patients on Earth.
	Preliminary data from our lab illustrates that Kasumi-3 myeloid progenitor cells (ATCC CRL-2725) are a viable and robust cell line for use in simulated microgravity (SMG) infection studies. We have validated the viability of these cells in SMG, validated an infection protocol using these cells and clinical-grade TB40/E cytomegalovirus (CMV) strain, and we have plated infected Kasumi-3 cells in both SMG and traditional FLASK conditions. Validation of Kasumi-3 in SMG: Kasumi-3 cells were brought up and maintained in ATCC® Roswell Park Memorial Institute (RPMI 1640) Medium, supplemented with 20% fetal bovine serum (FBS) and 1.4 mL of Penicillin-Streptomycin-Glutamine. Cultures were sustained in humidified 37°C /5%CO2 incubators, while all appropriate parameters for tissue culture were monitored throughout the entirety of the study. Cells were collected, combined, and divided among 4 culture vessels, two standard T-75cm2 canted neck flasks and two 50 mL rotating wall vessels (RWV), so that the plating density/concentration was 25 x 106 cells per vessel. We took aliquots from each SMG or FLASK culture vessel on days 1, 4, 8, and 12, to analyze cell count, cell size, and cell viability. Kasumi-3 viability increased immediately with exposure to SMG and was consistently higher throughout the trial with numbers ranging between 90-95%. Validation of Kasumi-3 infection protocol: Using the single vial of the TB40/E virus we obtained from Dr. Ritesh Tandon's lab, we passaged and propagated our own viral stock by infecting MRC5 human embryonic lung fibroblast cells. The "virus concentration or infectious dose" was determined by accomplishing a plaque-forming unit (PFU) assay where our value was 5.4x107 pfu/mL. We used the PFU to establish the multiplicity of infection (MOI) for the actual infection trial. We infected the Kasumi-3 cells from our SMG validation study above; from both conditions, in 12 well plates, MOI of 1, via high-speed centrifugation and then plated them in FLASKS with a plating density of 25 x 106 cells per vessel. Non-infected Kasumi-3 cells were maintained for use as our negative control also plated in the FLASK condition. We took aliquots from each FLASK on days 1, 4, 8, 12, and 16 to analyze cell count, cell size, and cell viability, and confirmed infection by RT-PCR. The data, shown below, show that cells pretreated to SMG have lower viral load 24 hours post-infection in 1g/FLASK condition. This suggests that perhaps the Kasumi-3 cells pretreated

Task Progress:	SMG are less susceptible to viral infection. We brought up fresh Kasumi-3 cells and maintained them in ATCC RPMI 1640 supplemented with 20% fetal bovine serum (FBS) and 1.4 mL of Penicillin-Streptomycin-Glutamine. Cultures were sustained in humidified 37°C /5%CO2 incubators, while all appropriate parameters for tissue culture were again monitored throughout the entirety of the study. Cells were maintained until concentrations conducive to large-scale infection were accomplished. Then cells were collected, combined, and then further divided (25 x 106) among SMG and FLASK conditions, as well as the non-infected controls for each gravitational group. Cells fated for infection were infected in 50 mL conical tubes at a MOI of 1 and then plated on 12 well plates. Non-infected cells were plated on separate 12 well plates, and then all plates were centrifuged at high speed. Cells collected from the 12 well plates were then plated in their respective vessels for their designated condition with a plating density of 25 x 106 cells per vessel. Non-infected Kasumi-3 cells were maintained for use as our negative control in both SMG and FLASK conditions. SMG was accomplished in 50 mL RWV rotating at 10 rpm. An aliquot was taken from each SMG or FLASK culture vessel on days 1, 4, 8, and 12. Cell count, cell size, and cell viability were obtained for all aliquots. Approximately 1 million cells were removed for RT-PCR analysis where viral load was determined for both cell pellet and supernatant fractions. We discovered that the cell pellet fractions for all gravitational conditions yield significantly higher viral load (ranged between 171 and 464, mean 318 ± SE 24) than supernatant fractions (<20 copies/ng of total DNA). We further determined that the viral loads from the FLASK/1g conditions were significantly higher than the SMG condition (p<.007). We, therefore, deduced that SMG impedes viral expansion probably due to the lack of direct cell contact, and we presented this data at the American Society for Gravitational and Space Research (ASGSR) conference as well as the NASA Human Research Program (HRP) workshop. In total, we have had six successful Kasumi-3 infection experiments. Each of these were confirmed via RT-PCR, but we additionally confirmed two of these infections using mab810X-Alexa 488(IE-1) antibody staining, and for one infection where we used a Towne-BAC-GFP CMV strain, we confirmed via confocal imaging. Radiation experiments: Kasumi cells that are hematopoietic stem cell-derived monocyte progenitor cells were infected with a low dose of CMV and establishment of latency was monitored by analyzing viral gene expression as well as genome copy numbers. Kasumi cells offer an established model of CMV latency and are amenable to latency establishment as well as reactivation in the laboratory. They do not allow lytic viral replication unless differentiated. Once CMV latency was established in these cells, the cells were exposed to gamma, proton, and GCR radiation (0.1, 0.25, 0.5, and 1.0 Gy) for different time points (e.g., 0 to 30 days) at the Brookhaven National Laboratory (BNL) NY. The cells were harvested at different times after the irradiation and analyzed for viral load by real-time PCR. It was interesting to note that there was an increase in the viral load for almost all the radiation concentrations used six days after the exposure of gamma, proton, and GCR. The data is being analyzed for statistical significance and interpretation. Genomics and Transcriptomics: Kasumi cells from the above experiments were also subjected to genomics and transcriptomics studies to reveal single nucleotide polymorphisms or other variations in the genome of irradiated cells compared to mock, and reveal any transcripts that are significantly altered in expression. Total nucleic acid (TNA) was extracted from the Kasumi cells from the above experiment and was sent for DNA as well as RNA sequencing at the Microbial genome sequence center (MiGS) at Pittsburgh, PA. Selected samples were sequenced from the above experiment. Genomic analysis: Initial set of 22 samples (pre-radiation, D6, D12, GCR, Proton) did not reveal any genomic variants. The analysis is still ongoing. Transcriptomic analysis: Interestingly, only one viral gene shows a significantly elevated level in GCR-Day6 cells compared to mock.
Bibliography Type:	Description: (Last Updated: 11/21/2025)
Articles in Peer-reviewed Journals	Mitra D, Hasan MH, Bates JT, Bierdeman MA, Ederer DR, Parmar RC, Fassero LA, Liang Q, Qiu H, Tiwari V, Zhang F, Linhardt RJ, Sharp JS, Wang L, Tandon R. "The degree of polymerization and sulfation patterns in heparan sulfate are critical determinants of cytomegalovirus entry into host cells." PLoS Pathog. 2021 Aug 5;17(8):e1009803. https://doi.org/10.1371/journal.ppat.1009803 ; PubMed PMID: 34352038; PubMed Central PMCID: PMC8384199. , Aug-2021
Articles in Peer-reviewed Journals	Pramanik A, Gao Y, Patibandla S, Mitra D, McCandless MG, Fassero LA, Gates K, Tandon R, Ray PC. "Aptamer conjugated gold nanostar-based distance-dependent nanoparticle surface energy transfer spectroscopy for ultrasensitive detection and inactivation of corona virus." J Phys Chem Lett. 2021 Mar 4;12(8):2166-71. https://doi.org/10.1021/acs.jpclett.0c03570 ; PubMed PMID: 33629859; PubMed Central PMCID: PMC7927280. , Mar-2021
Articles in Peer-reviewed Journals	Pramanik A, Gao Y, Patibandla S, Mitra D, McCandless MG, Fassero LA, Gates K, Tandon R, Ray PC. "The rapid diagnosis and effective inhibition of coronavirus using spike antibody attached gold nanoparticles." Nanoscale Adv. 2021 Mar 21;3(6):1588-96. https://doi.org/10.1039/D0NA01007C ; PubMed PMID: 34381960; PubMed Central PMCID: PMC8323809 , Mar-2021